RootAccess Overnight Sustainability Hackathon 2026

Rasayanam

Diwali should light up homes, not fill lungs.

"AI-powered air pollution prediction and health advisory system"

Team Higure_evilman

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One night a year, a city holds its breath

488

µg/m³ of PM2.5: Delhi's average air after Diwali 2025

8×

India's safe limit of 60 µg/m³

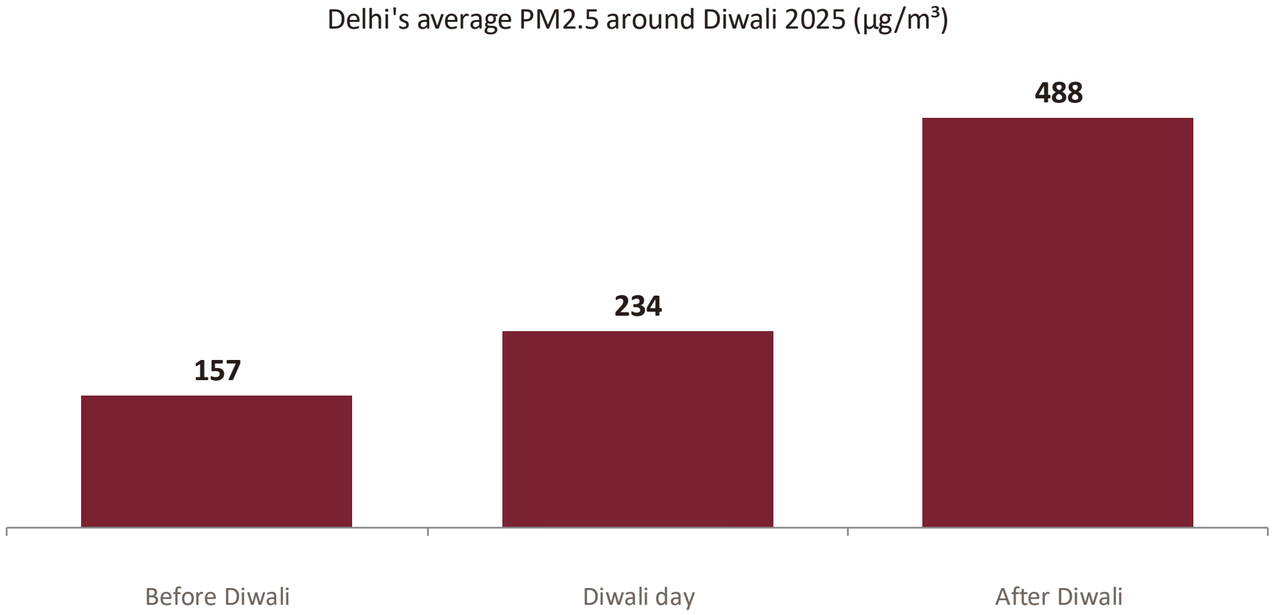
≈ 22

cigarettes a day, just by breathing

Sources: CPCB data, Climate Trends analysis (Oct 2025); cigarette equivalence from Berkeley Earth (22 µg/m³ for a day ≈ 1 cigarette).

Imagine Aarav. He is 8, has asthma, and lives near a busy Delhi crossing.

On Diwali night his parents must guess: shut the windows now or later? School tomorrow? A walk at noon? Every AQI app tells them only once the smoke is already in his lungs.



The forecast that arrives before the smoke

Rasayanam predicts PM2.5 and AQI for each monitoring station up to 7 days ahead, before and turns every prediction into advice a family can act on. Live, and updated every 10 minutes.

Aarav's Diwali, with Rasaynam

Day before, 6 pm

The 48-hour forecast shows his area turning severe from 9 pm on Diwali night.

Diwali, 8 pm

Windows shut, inhaler ready, breathing exercises from the Health Corner.

Next morning

The forecast shows a cleaner window at 2 pm. That's when he goes out.

Live AQI dashboard

AQI gauge, station selector and every pollutant, refreshed live.

7 days forecast

Hour-by-hour predictions plus Diwali climatology from past years.

Health Corner

Breathing exercises, care guidance, and when to see a chest or ENT doctor.

Learn

AQI and respiratory health, explained in plain words.

Technical approach

1 Data

- Historical CSV, 2020–2025
- 46,848 hourly rows, 8 stations
- Weather: temperature, humidity, wind, pressure, rain
- Pollutants: PM2.5, PM10, NO_2, SO_2, CO

2 Machine learning

- Cleaning, station-wise sorting
- Lags, rolling means, seasonality, Diwali proximity
- Target: CPCB AQI and PM2.5
- HistGradientBoosting, 3 h and 24 h models

3 Backend

- FastAPI + Uvicorn
- 48-hour recursive forecast: 16 steps per station
- CPCB sub-index AQI and health advisory
- REST /api and WebSocket /ws, every 10 min

4 Frontend

- HTML, CSS, JavaScript, Chart.js
- Live dashboard, PM reports
- Forecast and Diwali trends
- Health Corner and Learn

Honest test

Trained on 2020–2024, tested on 2025, a year the model never saw.

Live by design

WebSocket pushes fresh data every 10 minutes and reconnects on its own.

Reproducible

Model, metrics and feature spec saved as files and loaded at start-up.

Impacts and benefits

Families

Plan around the smoke: when to shut windows, when it's safe to step out.

Asthma and heart patients

Advance warning to keep medicines ready and avoid the worst hours.

Hospitals

See severe nights coming and prepare for breathing emergencies.

City authorities

Act before limits are crossed, not after the headlines.

From “the air is bad” to “the air will be bad at 9 pm, so here is what to do.”

That shift, from reacting to preparing, is what protects a child's lungs.

Goals we serve

SDG 3 Good health and well-being

SDG 11 Sustainable cities and communities

SDG 13 Climate action

Diwali 2026 is on 8 November.

Let this one be about the lights.

Rasayanam forecasts the air before it turns, and tells every family what to do.



Thank you

Team Higure_evilman | github.com/kartikeyag-13/Rasaynam